ECE 501 DIGITAL IMAGE PROCESSING
SAR Optical Feature Fusion For Crop Type Classification
REPORT WEEK 5 GROUP 11

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1. Background/Theory

This project has the objective of estimating yield using satellite information. We use YieldSAT benchmark dataset, which has several countries of field-level metadata (centroid, sowing date, harvest date) and yield data. This report represents 1,302 fields:
Argentina: 751 fields, Brazil: 551 fields. The optical sensors detect the reflected sunlight in the visible and near-infrared bands. Near-infrared light is strongly reflected by healthy vegetation and is absorbed by unhealthy vegetation. This is used in the Normalized Difference Vegetation Index (NDVI) which ranges from −1 to 1. Low values are provided by bare soil and water; High values are provided by dense green canopy. Optical data is limited by cloud cover which is a major challenge when capturing growth in the growing season of tropical and subtropical regions, and is widely used to gauge crop growth. SAR is an active sensor: It sends a set of microwaves and receives the signal scattered back off the surface. Since microwaves can penetrate clouds, they can be used to observe all weather and at night via SAR. Sentinel-1 works in C-band, and collects data in Interferometric Wide (IW) swaths with VV and VH polarizations at approximately 10 m pixel spacing in the Ground Range Detected (GRD) product. Backscatter typically is reported in dB (decibels). The combination of the two channels, VV and VH, is primarily sensitive to soil and canopy structure and primarily represents crop development through the season on crops, whereas VH is highly sensitive to the volume scattering of vegetation.

Formula used for NDVI:
NDVI = (NIR -Red) / (NIR + Red)

Why does co-registration matter?

In time series analysis the pixel on each date corresponds to the same spatial point at the ground. Any small movements in the geometric registration of acquisitions (even a fraction of pixel) introduces noise into the temporal signal. It estimates and corrects the shifts. With

a single relative orbit per field, the geometry of the view remains the same so as to minimize the variation (other than the crop) in the viewed region.

2. Materials and Tools

- YieldSAT raw archive: metadata JSON files per field (centroid latitude/longitude, sowing date and harvest date).
- Sentinel-1 GRD: IW mode, VV + VH, converted to dB, Sowing to harvest of each field.
- Data and Python: [list libraries, e.g. pandas, numpy, geopandas, matplotlib]
- Optical data with [e.g. Sentinel-2 or the optical bands of YieldSAT, and the bands used]
- SDTS for the data interchange between the various applications.
- QGIS with visual inspection of SAR tiles and centroids.

SCRIPT	PURPOSE	OUTPUT
inventory.py	Read raw json metadata	Inventory_argetin_brazil.csv, raw_json_centroids.csv
download.py	Download Sentinel-1 GeoTiffs per field	download_log.csv
Co-registration	Align acquisition dates	coregistration_log.csv
makecsv.py	Tabulate SAR tile information	sar_tiles.csv
makefootprints.py	Create the tile footprints	sar_tiles.geojson
summary.py	Run for quality checks	verfication_report.csv

3. Methodology

RAW JSON INVENTORY

Results were read directly from the raw YieldSAT zip, with version control and sharing accomplished via GitHub. We recorded the raw centroid latitude/longitude for all 1,302 fields and sowing dates and harvest dates. Also, using raw centroids instead of the preprocessed CSVs resolved an earlier problem of empty geometries.

SENTINEL 1

5 sets of Sentinel-1 GRD scenes (IW, VV + VH in dB) were requested for each field with one relative orbit per field, from sowing to harvest. It ended up producing a multi-date GeoTIFF file around the centroid in the local location. The average file size is approximately 4.8MB, just below the 5MB goal. If any fields did not have any VV + VH acquisition point in their season, it was noted in the download log.

CO-REGISTRATION

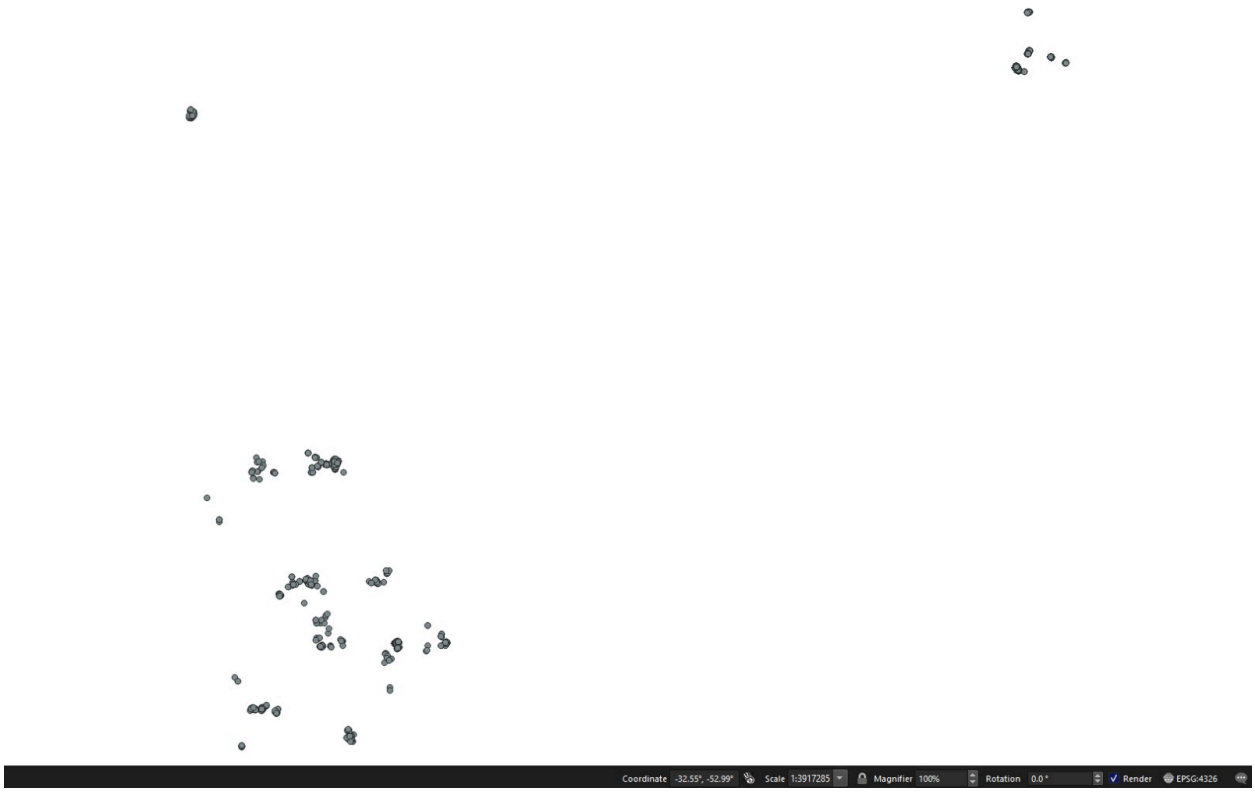
The dates of each field were co-registered to each other. The move for each date was determined and applied when the estimate was accurate. Installs where a reliable shift measurement was not possible were not changed.

VERIFICATION

We confirmed that all the raw centroids are within their SAR tile and computed the distance between their raw centroids and the centre of their SAR tile. The fraction of valid pixels and backscatter range were automatically checked as well. The processing of a visual check in QGIS (raw centroids and SAR tiles) is underway.

4. Results:





STEP	**RESULT**
Field in inventory	1302(751 Argentina, 551 Brazil)
Fields with Sentinel-1 VV +VH Data	1,235(94.9%) 67 without data is in Argentina
Fields co-registered	1,228
Dates Corrected	88.1%, median shift is 3.2 m
Raw centroids inside SAR tile	1,228 out of 1,228 median offset is 2.8m and maximum is 7.4m
Fields passing all automatic checks	1,206(98.2%), 90% valid pixels
Median backscatter	-11dB(VV), -18dB(VH)

5. Discussion:

Optical versus SAR

The optical analysis reveals the changes in the crop condition during the season, However, it is reliant on cloud free acquisition of the data. SAR data are so extensive that they span nearly all the fields (94.9%) and are weather independent and should complement the gaps in optical data.

Quality of the SAR pipeline

The small co-registration shift of the series (3.2 m) confirms that the series was already well aligned, while the typical backscatter values (around −11 dB VV, and around −18 dB VH) are within the range that is expected for cropland. An automatic check has been performed and 98.2% of the data passed, meaning there is consistency in the data.

6. Limitations:

- **7 fields were not accounted**: from1235 fields downloaded, 1228 were co-registered. The reason should be recorded in the co-registration log.
- **The centroid test being soft test**: Tiles are constructed around centroids which means that if the centroid falls within a tile, then it is fairly likely. The relevant field polygons are checked in QGIS, it is meaningful validation.
- **Maximum offset of 7.4 m**: The difference should be explained (reprojection/rounding) since pixel snapping should result in approximately 7.07m on a 10m grid.
- **Bias due to missing data**: The country balance (751 to 684 fields) is also altered by the fields without SAR data that are all located in Argentina. There are 22 fields which failed the checks that still require a decision – drop, flag or re-download.
- **Reproducibility**: GeoTIFFs are only stored locally, making full reproduction of the results in the repository impossible.

7. Learnings:

- The geometry can be retrieved directly from the raw JSON files and will not contain any errors from the preprocessed tables.
- By restricting the viewing geometry to one relative orbit per field, the time series is made simpler.
- All steps are captured (downloading, co-registration and verification), allowing easy tracing of why fields are missing.

- The strength of a check is dependent on its independence from the processing. In addition to automatic checks, a validation with an independent geometry is required.

8. Next Steps:

- Finish the QGIS visual check and add the screenshots.
  7.4 m maximum offset
- Explain the 7 missing co-registered fields.
- Determine what to do with the 22 failed fields and 67 missing data fields.
- Update the readme and place co-registration script into the repository.

9. References:

1. Torres, R., et al. (2012). GMES Sentinel-1 mission. *Remote Sensing of Environment*, 120, 9–24.
2. Rouse, J. W., Haas, R. H., Schell, J. A., & Deering, D. W. (1974). Monitoring vegetation systems in the Great Plains with ERTS. *Third ERTS Symposium, NASA SP-351*, 309–317.
3. Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2), 127–150.
4. YieldSAT: A Multimodal Benchmark Dataset for High-Resolution Crop Yield Prediction, 2026, arXiv:2604.00940. [Online]. Available: https://arxiv.org/abs/2604.00940. Dataset page: https://yieldsat.github.io/
5. ESA Sentinel-1 User Guide: https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-1-sar